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$$\begin{aligned} & \lim_{x \rightarrow +\infty} \frac{\sqrt{4x^2 - 5x + 1}}{-3x - 7} \\ &= \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 \left(4 - \frac{5}{x} + \frac{1}{x^2}\right)}}{-3x - 7} \\ &= \lim_{x \rightarrow +\infty} \frac{2x}{-3x - 7} = -\frac{2}{3} \\ & \lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 - 5x + 1}}{-3x - 7} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 \left(4 - \frac{5}{x} + \frac{1}{x^2}\right)}}{-3x - 7} \\ &= \lim_{x \rightarrow -\infty} \frac{-2x}{-3x - 7} = \frac{2}{3} \end{aligned}$$

## References